
ARTICLE DRAFT

Why BI Is Not Enough: The Move Toward Decision Intelligence

Promo-pricing, client analytics, and the new management landscape

Promotions are not discounts. They are control mechanisms.

Business intelligence made retail management more transparent. But transparency is not the same as control. In the personalization era, promotion management is no longer only about reading reports and choosing discounts. It is about coordinating customer-level interventions, category economics, loyalty mechanics, budgets, and future behavior.

This article develops that argument through the promo-pricing lens, using the attached retail-store case study as the main illustration. External side-paper quotations are intentionally omitted.

Core thesis

BI remains necessary, but it is no longer sufficient. A dashboard can explain what happened; a decision-intelligence system can recommend, constrain, execute, and learn from the next best intervention.

The management loop is changing

The classic loop is simple: the manager asks BI for information, interprets the dashboard, designs the action, executes it, and returns to the dashboard to see what happened. That loop worked when decisions were aggregated, slow-moving, and relatively independent.

Promo-pricing is moving in the opposite direction. It is becoming customer-specific, dynamic, and connected across categories. The decision space is too large for manual interpretation alone. The manager still owns the goal, the trade-off, and the accountability. But the operational formation of the solution increasingly belongs to a decision engine that combines client analytics, AI, and formal optimization.

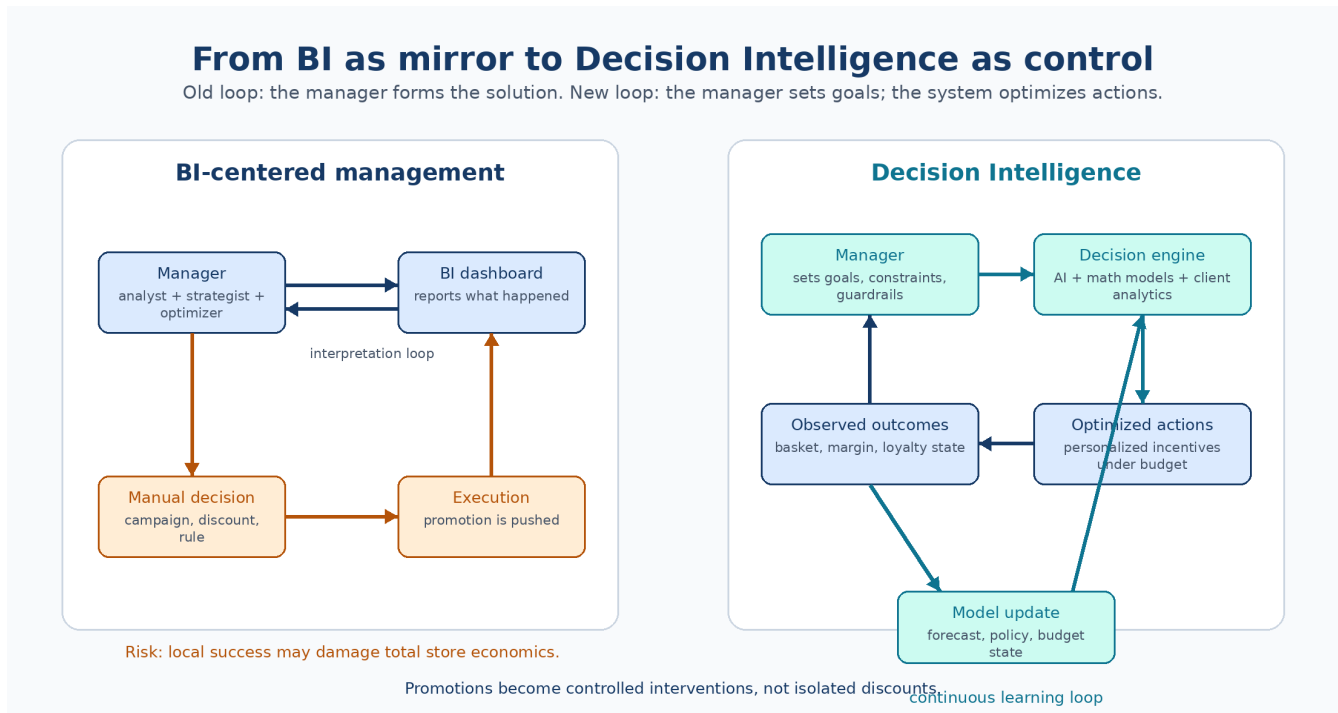


Figure 1. The move from a BI-centered interpretation loop to a decision-intelligence control loop.

1. Promotions are control mechanisms

For decades, retailers treated promotions as commercial instruments: reduce the price, stimulate demand, defend market share, support supplier funding, clear stock, or bring traffic into the store. In that view, the discount is the action and BI is the mirror. Managers review sales, margin, conversion, and basket dynamics, then use experience to decide the next promotion.

That model is no longer enough. In a personalization context, a promotion is a behavioral signal sent to a specific customer or segment, at a specific moment, under specific budget and margin constraints. The same discount can increase total spend for one customer, cannibalize another category for another customer, accelerate stock depletion, damage margin, or shift loyalty costs into the future.

A promotion therefore should not be understood as an isolated price cut. It is a control variable inside a dynamic retail system. The real question is not only whether a campaign performed well after the fact. The real question is which intervention should be applied now, to whom, and under what constraints, so that the whole store moves toward the target state.

Old question: Which promotion performed better?
New question: Which intervention should the system apply next?

2. The BI loop reaches its limit

BI is excellent at visibility. It can show that a promotion increased sales in a category, that margin dropped, that basket composition changed, or that a customer cohort responded better than another. These insights are valuable. They are also incomplete when the organization needs to choose the next action in a complex system.

The gap appears when several conditions hold at the same time: categories have separate targets, customers move across different purchase states, promotions influence behavior beyond the promoted SKU, and loyalty incentives operate under finite budgets. In that environment, a dashboard can reveal tension, but it does not automatically resolve it.

For example, Category A may show strong promotional uplift while Category B loses spend because customers switch or reallocate their budget. A BI report can display both facts. It still leaves the manager with the harder problem: should the retailer accept local category success if it damages the total store objective? Should the loyalty program compensate? Which customers should receive the compensating incentive? How deep should it be? How much budget should be held for tomorrow?

These are not reporting questions. They are decision questions. More precisely, they are prediction, optimization, and control questions.

3. Category management creates local intelligence, not system intelligence

Retail organizations are often built around categories. This creates ownership and commercial clarity: each category manager knows suppliers, competitors, seasonality, product assortment, and price sensitivities. That knowledge remains important. But it is local knowledge.

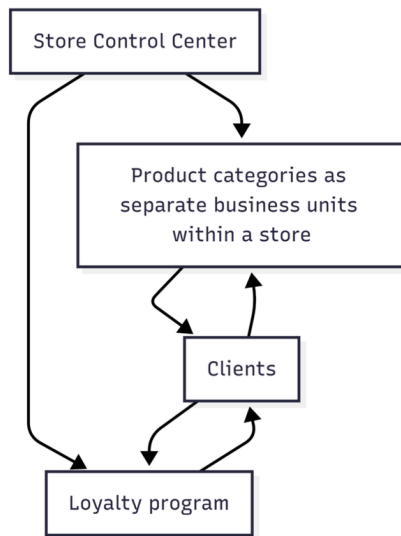
The structural risk is that each category can optimize its own metrics while the store loses at the system level. A manager may choose a promotion that improves a category dashboard and still creates a negative cross-category effect. The attached case study frames exactly this tension: product categories act as separate business units, while customer behavior connects them through baskets, budgets, and switching effects.

The implication is not that category managers are weak. The implication is that the decision problem has outgrown the old operating model. No local manager can manually calculate, in real time, the full effect of thousands or millions of personalized interventions across customers, categories, budgets, loyalty states, and future revenue expectations.

The organization needs system intelligence on top of local intelligence.

Problem solution framing

Stochastic game between three levels



The problem of managing the disbalance between category-level and store-level objectives can be formally defined as a stochastic setting where the Center does not control agent actions directly, so the Center chooses control actions to respond to agents actions in multi-stage game (Bellman) in order to optimize its objectives.

$$\max_{u_t \in \mathcal{U}} \mathbb{E}_{a_t \sim \mathcal{P}_t} [R_{\text{global}}(a_t, u_{t-1}) + \gamma \cdot V_{t+1}(s_{t+1})]$$

where:

- a_t - agents' (categories) actions at time t , drawn from $\mathcal{P}_t$,
- u_{t-1} - Center's response chosen at previous step $t - 1$,
- $R_{\text{global}}(a_t, u_{t-1})$ - revenue at time t , depending on a_t and prior Center response,
- s_{t+1} - system state after actions (a_t, u_t) ,
- $V_{t+1}(s_{t+1})$ - expected future revenue starting from next state,
- γ - discount factor (optional).

Figure 2. Case illustration: the store control center, category agents, clients, and loyalty program form a multi-level decision system. Source: attached DFDE presentation, page 5.

4. Client analytics changes the nature of promo-pricing

Traditional BI aggregates customers into reports: segments, regions, cohorts, average baskets, response rates, and purchase frequency. Client analytics goes further. It tries to understand behavioral trajectories: what a customer is likely to buy, how they respond to incentives, which categories they combine, when they are price-sensitive, and how their loyalty state changes over time.

Once promotion management becomes customer-level or segment-level, the retailer is no longer simply choosing a campaign. It is managing a flow of customers through states. A loyalty program is no longer only a marketing tool. It becomes a control layer that can regulate who receives what incentive, when, and at what financial cost.

This is why personalization makes BI insufficient. Personalization increases the number of possible actions. It also increases the number of constraints: budget, margin, fairness, customer experience, category balance, and long-term value. The manager can define these priorities, but the system must calculate feasible actions at scale.

Problem solution framing

$$x_{i+1}(t+1) = x_{i+1}(t) + u_i(t) \cdot \mu_i x_i(t) - D_{i+1}(t), \quad (1)$$

where $x_i(t)$ denotes the number of customers at level A_i ; $u_i(t) \in \{0, 1\}$ is a control variable indicating admission to the next level; μ_i is the progression rate (the scaling factor determining the size of the delta from $x_i(t)$); $D_{i+1}(t)$ captures departures from level A_{i+1} . The model enforces capacity constraints:

$$x_i(t) \leq C_i, \quad \forall i, t, \quad (2)$$

which effectively block further transitions when a level is full. This blocking mechanism accounts for additional economic information about the system state and serves as the control lever to regulate customer flow and prevent over-discounting.

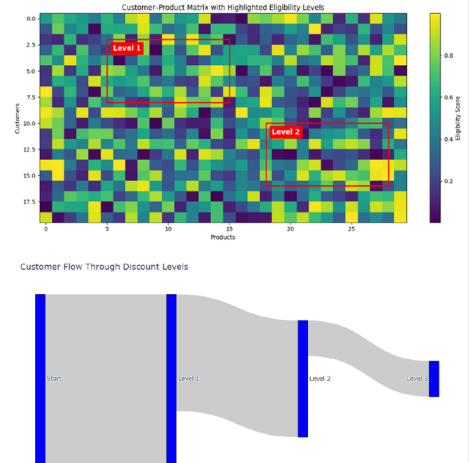


Figure 3. Case illustration: the loyalty system can be treated as a managed flow with levels, capacity constraints, and control variables. Source: attached DFDE presentation, page 6.

5. From pushing to steering

A short-horizon promotion strategy often pushes discounts into the market. The retailer applies incentives to generate immediate response, then checks the result after the fact. This is tactically useful, but it can become expensive, reactive, and poorly coordinated.

Steering is different. Steering means the retailer uses promotions to guide customer behavior toward desired system outcomes: better total revenue, healthier category balance, controlled loyalty costs, and more efficient promotional spend. The intervention is no longer judged only by immediate uplift; it is judged by how it moves the system under constraints.

The case study expresses this idea through a model predictive control logic. Managers set category prices. The system estimates whether those actions create revenue damage. If the damage is significant, it calculates the compensating discount policy, applies the optimal intervention, updates the loyalty state, records actual revenue, and uses the new result to train the model.

Model Predictive Control approach to solution

Algorithm 1 Price Damage Compensation

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1: Initialization: Load current state (customers, finances, price history)
2: for every day  $t = 0, 1, 2, \dots, T$  do
3:   Step 1: Managers set prices  $a_t$ 
4:   Step 2: Estimate damage:  $\Delta R = f(a_t, \text{history})$  ▷ Predict revenue loss
5:   if  $\Delta R < -\epsilon$  then                                ▷ If damage is significant
6:     Step 3: Optimize discounts:

       
$$\min_{u_t} |R^{\text{target}} - R(a_t, u_t)| + \lambda \sum u_i x_i$$


       Discounts should not cost more than they compensate

7:   Step 4: Apply optimal  $u_t^*$ 
8:   else
9:     Step 3-4: Use standard discounts
10:  end if
11:  Step 5: Update loyalty:  $x_{t+1} \leftarrow F(x_t, u_t, a_t)$ 
12:  Step 6: Record the result: Record the actual revenue  $R_t$ 
13:  Step 7: Train the model: Update the damage forecast on new data
14: end for

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Figure 4. Case illustration: a model predictive control workflow for price-damage compensation. Source: attached DFDE presentation, page 7.

The phrase "model predictive control" matters because it captures the shift from passive analytics to active management. The system does not merely describe the last campaign. It forecasts the effect of actions, compares them against a target, chooses a policy under constraints, observes the result, and updates itself.

In commercial language: the retailer moves from pushing discounts to steering behavior.

Further research: practical implementation

Short-horizon solution

Pushing

VS

Sequential User-behavior modelling

Steering

Figure 5. Case illustration: short-horizon pushing versus sequential user-behavior steering. Source: attached DFDE presentation, page 9.

6. What the decision engine actually does

The decision engine is not magic. It is a structured layer between business goals and operational actions. It receives the current state of the system, such as customers, finances, price history, category actions, and loyalty states. It evaluates the gap between the target and the likely result. Then it searches for an intervention that improves the objective without breaking constraints.

In promo-pricing, the engine needs to answer questions such as:

- Which customers or segments should receive an incentive?
- Which categories should be stimulated, protected, or left untouched?
- How deep can the incentive be before it costs more than it compensates?
- What is the expected effect on total revenue, not only on one category?
- How should the model update after the observed result?

This is where proven mathematical models become operationally important. They translate managerial intention into feasible decisions. They also make the trade-off explicit: revenue gap versus intervention cost, local category target versus store-level objective, immediate response versus future value.

Step 3 (Optimize discounts) specification

$$\begin{aligned}
 \min_u \quad & \underbrace{\left| R^{\text{target}} - \sum_{k=1}^K p_k(a_t) \cdot q_k(u_t) \right|}_{\text{Term 1: Revenue Gap Penalty}} + \lambda \underbrace{\sum_{i=1}^L u_i x_i}_{\text{Term 2: Total Discount Cost}} \quad // \text{Min. cost of intervention} \\
 \text{s.t.} \quad & q_k(u_t) = \underbrace{d_k^0}_{\text{Baseline demand}} \left(1 + \gamma \underbrace{\sum_{i=1}^L u_i \cdot x_i}_{\text{Total Loyalty Stimulus}} \right) \quad // \text{Demand boost from overall program satisfaction} \\
 & 0 \leq u_i \leq 1, \quad // \text{Discounts are rates between 0\% and 100\%} \\
 & \sum u_i x_i \leq \text{Budget}_t \quad // \text{Hard constraint: Do not exceed the budget}
 \end{aligned}$$

Figure 6. Case illustration: optimization of discount intervention cost against a revenue target and budget constraint.
Source: attached DFDE presentation, page 8.

7. What changes for managers

The move from BI to decision intelligence does not remove the manager. It changes the manager's role.

In the BI-centered model, the manager is expected to interpret information and manually construct the answer. The manager looks at the dashboard, identifies the problem, designs the promotion, estimates the risk, and decides the action.

In the decision-intelligence model, the manager increasingly defines objectives, constraints, and strategic boundaries. The system forms and evaluates the operational solution. The manager chooses the path; the model calculates the route.

The new managerial task

The manager specifies the business goal, acceptable constraints, and governance rules: protect margin, recover revenue loss, avoid over-discounting, improve basket value, preserve customer experience, support strategic categories, or limit loyalty spend. The decision system then searches for the best feasible action inside those boundaries.

This is not "AI replacing managers." It is a shift from manual decision construction to goal-driven decision supervision. Human judgment moves upward: from calculating each operational answer to designing the system of objectives, limits, and accountability.

8. Why BI is still necessary but no longer sufficient

BI remains the foundation. Without clean data, trusted metrics, reporting discipline, and business transparency, decision intelligence has nothing reliable to act on. The argument is not that BI should disappear. The argument is that BI should no longer be treated as the final layer of the management system.

In personalized promo-pricing, the organization needs five capabilities beyond classic BI:

- Prediction: estimate what will happen if a customer receives a specific offer.
- Optimization: choose the offer that best supports the objective under budget, margin, and category constraints.
- Coordination: prevent one category's local success from damaging total store performance.
- Control: adjust interventions continuously as customer behavior and system state change.
- Learning: update the model after every action and observed result.

BI can support each of these, but it does not perform them by itself. That is the reason for the move toward decision intelligence.

9. The new management landscape

The management landscape is changing because the commercial decision itself is changing. In the old landscape, decision authority was closely tied to human interpretation. The best manager was the one who could read the numbers, understand the market, and make the right call.

In the new landscape, the best organization will be the one that combines human strategic judgment with formalized decision systems. Managers still define priorities, evaluate trade-offs, and carry accountability. But the operational formation of many decisions moves into AI-supported mathematical models.

This is especially true in promotion management, where decisions are frequent, interconnected, and increasingly customer-specific. A promotion is no longer just a commercial action. It is a system intervention. A discount is no longer just a price reduction. It is a control variable. A loyalty program is no longer just a retention mechanism. It is a way to regulate customer flow.

BI helped retailers see the business more clearly. But seeing is not the same as deciding.

Decision intelligence is the layer that turns information into coordinated action.

Closing

As promo-pricing becomes more personalized, dynamic, and system-dependent, the main challenge is no longer the absence of information. The challenge is converting information into coordinated, constrained, and continuously updated action.

That is why BI is not enough. Retailers need decision intelligence: a management approach that connects data, client analytics, mathematical models, AI, constraints, and business goals into one continuous decision process.

Promotions are not discounts. They are control mechanisms. In the personalization era, those mechanisms cannot rely only on dashboards and local managerial intuition. They require a new operating model in which managers set goals and strategic boundaries, while intelligent systems generate, test, optimize, and learn from the actions that move the business forward.

Source note

Figures 2-6 are adapted from the attached DFDE 2025 presentation, "Mathematical methods of managing a retail store as a dynamic system: the promotional pricing aspect," by Vladimir Belolipetskiy. External literature quotations have been intentionally omitted in this draft.